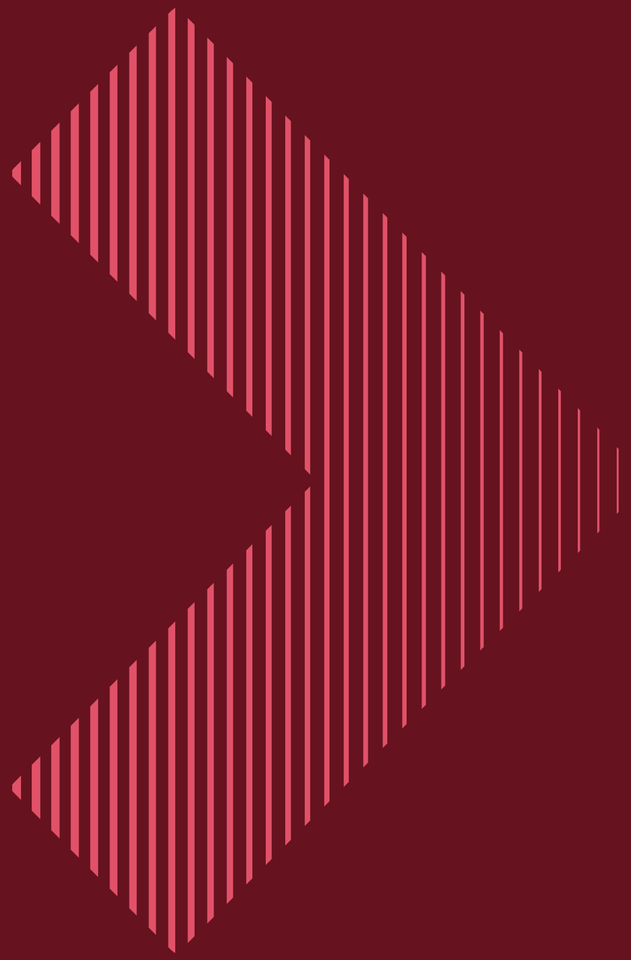
Central Bank Digital Currency:
Central Banking For All?

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CENTRAL BANKING POLICIES FOR SUSTAINABILITY

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Introduction

The technological innovations in cryptography and distributed ledger technology have made it possible to issue digital currencies. The adoption of a CBDC is not a purely technical issue: it represents a potential revolution in the architecture of the financial system.

The CBDC Debate

What is CBDC?

Barrdear & Kumhof (2016): "universal, electronic, 24x7, national-currency-denominated and interest-bearing access to central bank's balance sheet."

→ CBDC = consumers holding accounts directly at the central bank.

Key Research Question

- **What are the implications of a CBDC for maturity transformation and financial stability?**
- **Can a central bank 'open to all' replicate the efficiency of private financial intermediation, and how does it affect bank runs?**

Private vs. Public Digital Money



- **Private:** Bitcoin, Ethereum, Libra — triggered the CBDC debate
- **Public:** Central bank initiatives: Sveriges Riksbank (e-Krona), BIS position papers, Bank of Japan legal analysis
- **Political:** Swiss Vollgeld initiative (2018) rejected, but debate continues.



Research Questions & Methodology

Q1	• Will a CBDC impair the role of the financial system in allocating funds to productive projects?
Q2	• Can we reorganize the financial system so that a CBDC still allows the optimal flow of funds?
Q3	• Will a CBDC make bank runs disappear and stabilize the financial system?
Q4	• What happens to competition between the central bank and private intermediaries?
Methodology	• **Diamond & Dybvig (1983) model of maturity transformation, augmented with (i) a distinction between commercial and investment banks, and (ii) a government-controlled central bank offering CBDC accounts**

Historical Background: Central Banks as Retail Lenders

Bank of England (est. 1694)

Established as a privately owned limited-liability corporation with the right to maximize profits through issuing paper money, taking deposits, lending on mortgages and dealing in bills of exchange. Pursued commercial activities vigorously for over two centuries (Kynaston, 2017).

First & Second Banks of the U.S.

Both participated actively in borrowing and credit markets (Cowen, 2000; Knodell, 2017). The 'Bank War' between Andrew Jackson and Nicholas Biddle was directly linked to Second Bank operations with firms and merchants (Remini, 2017).

Banco de España (est. 1782)

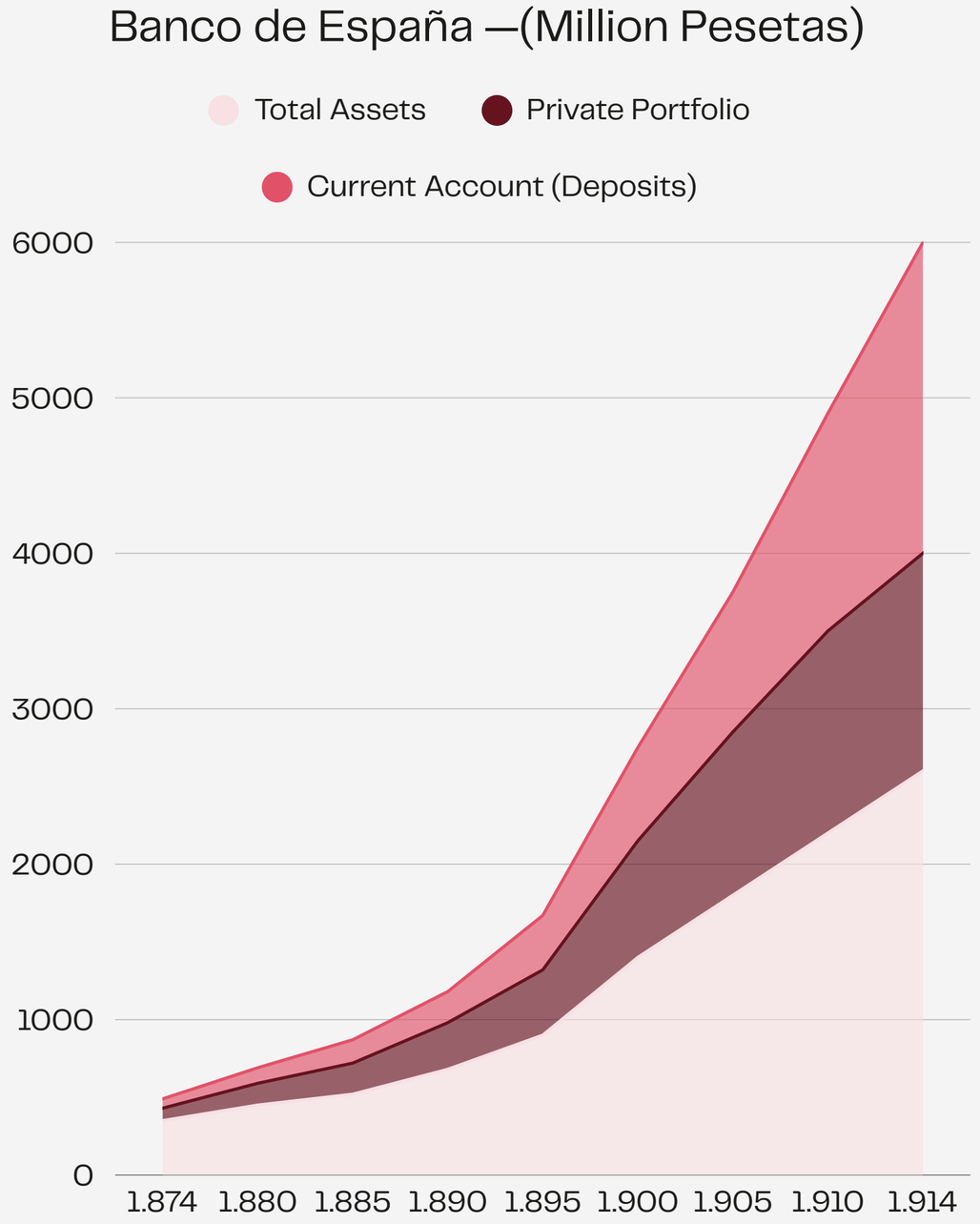
By 1900, held 68% of total assets and 73% of all demand deposits in the Spanish financial sector via 58 branches covering all major towns. Private loan portfolio exceeded public treasury bond portfolio by the early 20th century (Martín-Aceña, 2017).

Banco de España: Dominance in Financial Markets, 1874–1914

Private loan portfolio exceeded public treasury bond portfolio by start of 20th century — the central bank was primarily a commercial lender

- **68% of total financial sector assets held by Banco de España in 1900**

- 73% of all demand deposits in the Spanish financial sector in 1900



Source: Martín-Aceña (2017, 2018). Data are illustrative of the paper's Figure 1 trends

The Post-WWII Shift: Separation of Central Bank from the Public

1861	• UK Post Office Savings Bank opens — government-backed deposits for citizens via postal network
1911–1967	• U.S. Postal Savings System — reached ~10% of commercial banking assets by end of WWII
1946	• Bank of England nationalized (UK Labour government) — end of commercial banking ambitions
1962	• Banco de España nationalized (Franco regime) — separation from retail market
Post-WWII	• **Gold standard abolished → governments seek direct monetary policy control → retail activities transferred to commercial banks**
Digital Age	• **Internet eliminates operational barriers → CBDC makes central bank retail participation feasible again**

The Diamond–Dybvig model

t = 0

Period 0
Consumers endowed with 1 unit. Banks offer deposit contracts. Central bank offers CBDC accounts. All agents deposit.

t = 1

Period 1 (Interim)
Consumers learn their type (early/patient or late/patient). Withdrawal game is played. Bank runs may occur.

t = 2

Period 2
Long asset matures with return $R > 1$. Remaining depositors receive payoffs. Final consumption occurs.

(1) Commercial banks ≠ Investment banks | (2) Central bank competes for deposits via CBDC

A Diamond–Dybvig Framework with Investment Banks and a Central Bank

Consumers

Setup

- $[0,1]$-continuum of ex-ante identical consumers
- Each endowed with 1 unit of good at $t = 0$
- Access to storage technology (1 unit → 1 unit)
- Types revealed privately at $t = 1$

Utility function:

$U(c_1,c_2) = u(c_1)$ with prob. λ [early consumer]

$u(c_2)$ with prob. $1-\lambda$ [late consumer]

$u: \mathbb{R}_+ \to \mathbb{R}_+$ strictly increasing, strictly concave, C^1

Consumer Types

prob. λ

Early (Impatient)
Needs liquidity at $t=1$. Always withdraws in Period 1.

Consumer Types

prob. $1-\lambda$

Late (Patient)

Prefers $t=2$ consumption. May mimic early type during panic.

Informational Asymmetry

Each consumer's type is private information — banks cannot distinguish between early and late consumers, preventing type-contingent contracts. This asymmetry is the fundamental source of bank run vulnerability.

Commercial Banks & Deposit Contracts

Deposit Contract ($\hat{c}_1, \hat{c}_2$):

Consumer deposits 1 unit at t=0. Bank promises $\hat{c}_1$ on demand at t=1, or $\hat{c}_2$ (or available resources equally divided) at t=2.

Bank's Investment Decision

$y \in [0,1]$ fraction in short asset
$(1-y)$ fraction in long asset

Short asset returns:
Serves t=1 withdrawals: $\lambda\hat{c}_1 \leq y$

Long asset returns:
Serves t=2 payoff: $(1-\lambda)\hat{c}_2 \leq R(1-y) + y - \lambda\hat{c}$

Early liquidation:
Bank can liquidate long asset at $\kappa < 1$ per unit to meet excess t=1 demands

Optimal Contract (c^*_1, c^*_2)

Bertrand competition forces banks to maximize consumer utility → zero profit equilibrium.

First-order conditions:

$c^*_1 = y^*/\lambda$

$c^*_2 = R(1-y^*)/(1-\lambda)$

$u'(y^*/\lambda) = R \cdot u'(R(1-y^*)/(1-\lambda))$

Result: $c^*_1 < c^*_2$ (since $R > 1$ and u concave).
Ex-ante first-best efficient allocation.

Investment Banks

Definition & Role

Investment banks have access to both the storage and long-term technologies, but exclusively offer contracts with non-negative payoffs in period 2. **They do NOT provide demand deposits (no t=1 callable liabilities). All cash flows are determined at t=0; no spontaneous withdrawals are permitted.**

Contract with CB

$\ell_2 \in \mathbb{R}_+$

Real liabilities per unit deposited by central bank in t=0. IB commits to investing all received funds in long asset.

Profit constraint

$\ell_2 \leq R$

Investment bank only accepts contract if profitable (non-negative profit condition). Competition drives $\ell_2 \to R$.

Scope

Broad definition

Includes industrial banks (Continental Europe, Japan, Korea), retirement funds — any long-term return vehicle.

No callable claims

Non-callable

Central bank CANNOT demand early repayment. This rigidity is the key mechanism deterring central bank runs.

Model: The Central Bank

Contracts with Investment Banks

To access long-term returns, the central bank contracts via wholesale deposits with investment banks. **The loan $\ell_2 \leq R$ is non-callable — the key rigidity. CB already participates in wholesale markets** (repo operations), making this empirically plausible.

No Fiscal Backing

The central bank has no independent source of taxation and receives no direct or indirect fiscal subsidy. This assumption prevents trivial results: fiscal backing would give the CB an unfair competitive advantage over commercial banks.

Government-Controlled Institution

Control, not ownership, defines the central bank's behavior.

Access to Short-Term Asset Only

The central bank can store real goods (short asset), but lacks access to the long-term productive technology.

Equilibrium

Definition 1 (Symmetric Equilibrium) consists of

1	• A contract ℓ_2 between the central bank and the representative investment bank
2	• A demand-deposit contract $(\hat{c}_1, \hat{c}_2)$ for the representative commercial bank
3	• A demand-deposit contract (d_1, d_2) for the central bank
4	• Deposit decisions $h \in \{0,1\}$: consumers choose CB ($h=1$) or commercial bank ($h=0$)
5	• Strategy profile $\sigma \in \{1,2\}$ for the withdrawal game in the interim period
6	• Fraction $\alpha \in [0,1]$ of depositors who withdraw in period 1
7	• Fraction $f \in [0,1]$ of consumers depositing with the central bank

the equilibrium of the model is a set of contracts and strategies that satisfies simultaneously all the optimal conditions for all agents

Replication of the Optimal Contract

Lemma 4.1 — Replication of the Optimal Contract

The central bank replicates the socially optimal commercial bank contract by setting:

$x = y^* \cdot d_1 = c^*_1 \cdot d_2 = R(1-y^*) / (1-\lambda)$

To offer this optimal contract, the central bank requires $\ell_2 = R$, implying zero profits to investment banks.



The Equivalence Result

Proposition 4.1 — Equivalence of Financial Intermediation

In equilibrium, the socially optimal contract is offered either by the commercial banks, the central bank, or both.

f = 0

Only commercial banks have customers. Bertrand competitio ensures socially optimal contract via zero-profit equilibrium.

CB Absent

f ∈ (0,1)

Both compete. Indifference implies BOTH offer optimal contract. Any f is an equilibrium. Maturity transformation preserved.

Coexistence

f = 1

Central bank absorbs all deposits. Occurs when CB is perceived as safer — but monopoly ay endanger optimal transformation.

CB Monopoly

The Sinister Counterpart of the Equivalence Result

Critical Caveat

The equivalence result holds only when competitive forces operate freely. Disruptions to competition destroy the mechanisms ensuring optimal maturity transformation.

Fiscal Subsidization

If the central bank receives fiscal backing (explicit subsidies, seigniorage, tax advantages on CB deposits), commercial banks cannot compete on equal footing

Political Economy Distortions

Political groups will lobby a central bank 'open to all' to change lending and deposit policies for distributional goals. Without market discipline, the CB may deviate from efficiency toward politically motivated credit allocation — a form of financial repression.

Investment Bank Extraction

For any $\ell_2 < R$, the CB fails to offer the optimal contract, loses deposits to commercial banks, and investment in long assets collapses.

Propositions 5.2 & 5.3: No Run on the Central Bank

Prop. 5.2 — Punishment Regime

"Central bank denies payment to depositors beyond measure λ who withdraw during a run. All t=2 returns go exclusively to roll-over depositors."

Unique equilibrium: only impatient depositors withdraw (α = λ). Runs do not occur. Socially optimal contract always attained.

uKey: $\lambda/\alpha \cdot u(c^*_1) < u(R(1-y^*)/(1-\alpha))$ for $\alpha > \lambda$ → rolling over strictly dominates.

Prop. 5.3 — Equal Treatment Regime

Depositors who could not be served at t=1 are treated as if they had rolled over. Roll-over payoff becomes independent of others' actions.

Unique equilibrium: only impatient depositors withdraw (α = λ). Runs do not occur. Socially optimal contract always attained.

Key: $c^*_1 < c^*_2$ and roll-over payoff = c^*_2 regardless of α → rolling over strictly dominates

Intuition

The non-callable loan protects the long asset's return for roll-over depositors. Even if many withdraw, those who stay receive $R(1-y^*)/(1-\alpha) \to \infty$ as $\alpha \to 1$. Under both regimes, the payoff from rolling over always exceeds that from withdrawing for a patient depositor.

Proposition 5.4: The Central Bank as Deposit Monopolist

If the central bank offers the socially optimal contract, then independently of whether it punishes run-participants or applies equal treatment, it attracts ALL deposits away from the commercial banking sector (f = 1).

Why the Monopoly Arises

U_1 = expected utility from run-prone commercial bank contract

U_2 = expected utility from run-proof CBDC contract

Since commercial bank has positive run probability:
$U_1 < U_2$

The Double-Edged Sword

✓ **Benefit: Eliminates bank runs, financial stability improved**

✗ **Risk: CB can exploit monopoly to offer $c_1 < c^*_1$ while still attracting all deposits (since $U(c_1) > U_1$ even for $c_1 < c^*_1$)**
→ Suboptimal maturity transformation without competitive discipline

The resilience of central banks to runs is a double-edged sword: it avoids financial panics,

but it destroys the competitive forces that discipline central banks open to all."

Section 5: Bank Runs

Commercial Bank

- Can liquidate long asset at t=1
- (at cost κ < 1 per unit)

- • Two pure Nash equilibria:
- — Good: α=λ (only impatient withdraw)
- — Run: α=1 (all depositors panic)

- • Strategic complementarity:
- If others run → rational to run
- If no run → rational to roll over

- • Self-fulfilling prophecy — no fundamental basis required

Central Bank

- • **CANNOT** liquidate investment bank loan
- → Only serves up to x = y* from short asset

- Contract rigidity breaks strategic complementarity:
- — Under punishment: unique equilibrium, no run (Prop. 5.2)
- — Under equal treatment: unique equilibrium,
- no run (Prop. 5.3)
- • Depositors internalize stability → CB becomes deposit monopolist

Run Condition: $(\alpha-\lambda)c^*_1 > (1-y^*)\kappa$

Bank must liquidate long asset to cover excess t=1 withdrawals. If insufficient assets → rationing and zero payoff to late withdrawers.

Good Equilibrium (α = λ)

Run Equilibrium (α = 1)

Event/ Action	withdraw	roll-over
no run,	$u(c_1^*)$	$u\left(\frac{R[(1-y^*)-(\alpha-\lambda)c_1^*/\kappa]}{1-\alpha}\right)$
run	$\frac{y^*+(1-y^*)k}{\alpha c_1^*}\cdot u(c_1^*)$	0

Fundamental Runs Under Asset Risk

Extension: The long asset pays R with probability $p(\theta)$ where $\theta \sim U[0,1]$ is the state; depositors observe noisy private signals $\theta_i = \theta + \varepsilon_i$. This global games environment selects a unique equilibrium.

Commercial Bank

- Run equilibrium persists in GG setting
- Dominance regions define run threshold $\hat{\theta}$
- Strategic miscoordination broadens run region
- Runs occur for all states $\theta \in [0, \hat{\theta})$
- $\hat{\theta} > \bar{\theta}$ (wider run region than CB under equal treatment)

CB — Equal Treatment

- Every patient depositor is pivotal
- No coordination game structure
- Dominant action: withdraw if $\theta < \bar{\theta}$ (condition: $u(c^*_1) = p(\bar{\theta})u(R(1-y^*)/(1-\lambda))$)
- Roll over dominant for $\theta > \bar{\theta}$
- Runs occur for $\theta \in [0, \bar{\theta})$, but NOT for $(\bar{\theta}, 1)$

CB — Punishment

- Even stronger deterrence than equal treatment
- For $\alpha \to 1$: pro-rata payoff $\to \infty$ for roll-over
- Bounded withdraw payoff: $\lambda / \alpha \cdot u(c^*_1)$
- Rolling over optimal for all $\theta > 0$ and $\alpha > \alpha(\theta)$
- CB at least as stable as under equal treatment

Final Considerations

	Implcations	Conclusions
1	**CBDC is Feasible but Not Trivially Beneficial**	**MAIN RESULTS**: A CBDC (account-based, interest-bearing) allows the central bank to compete for deposits while relying on investment banks for long-term investment. Absent panics, the socially optimal allocation is attained — the equivalence result.
2	**No Fiscal Backing Is Essential**	Competition from private banks disciplines the CB and investment banks. Fiscal backing, political economy forces, or any impairment of commercial banking competition can lead to suboptimal maturity transformation.
3	**Regulate Against Maturity-Transforming CB Assets**	• **Run Deterrence**
4	**Beware the Monopoly Power Trap**	• **Deposit Monopoly**
5	**Efficiency-Stability Trade-off Under Asset Risk**	• **Extensions**